Email & SMS Spam Detection System

System Architecture Specification & Technical Documentation — Category: Natural Language Processing & Cybersecurity

Author: Rakshitha S (Full-Stack & AI/ML)

Institution: Jnana Vikas Inst of Tech, Bengaluru

Repository: https://github.com/rakshapatil526/Email-Spam-Detection-System

Status: Verified Repository

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1. Domain Problem & Objective

Spam communications represent a critical cybersecurity threat, delivering phishing exploits, trojans, and financial fraud. The objective of this project is to build an ultra-lightweight, explainable NLP classification engine that achieves 100% precision on legitimate messages (Ham) while catching deceptive promotional spam.

2. Preprocessing & NLP Pipeline

• **Text Cleaning:** Removal of special characters, currency symbols, and excessive punctuation.

• **Tokenization & Lemmatization:** Word splitting, stop-word removal, and morphological base reduction.

• **Vectorization:** TF-IDF vectorization mapping term frequency relative to inverse document frequency.

• **Classifier:** Multinomial Naive Bayes model tuned with Laplace smoothing (alpha=1.0).

3. System Modules & Tech Stack

• **Modules:** Single Message Predictor, Batch CSV Message Analyzer, and Model Performance Inspector.

• **Technologies:** Python, Streamlit, Scikit-learn, NLTK, Pandas, Matplotlib.

• **Status:** Fully verified from repository source code and unit tests.